

Exploratory Data Analysis

Introduction

This chapter will show you how to use visualisation and transformation to explore your data in a systematic way, a task that statisticians call exploratory data analysis, or EDA for short.

EDA is an iterative cycle. You:

1. Generate questions about your data
2. Search for answers by visualizing, transforming and modelling your data
3. Use what you learn to refine your questions and or generate new questions

Note

"There are no routine statistical questions, only questionable statistical routines." — Sir David Cox

Tip

"Far better an approximate answer to the right question, which is often vague, than an exact answer to the wrong question, which can always be made precise." — John Tukey

Important

Your goal during EDA is to develop an understanding of your data. The easiest way to do this is to use questions as tools to guide your investigation.

In EDA two types of questions exists namely:

1. What type of variation occurs within my variables?
2. What type of covariation occurs between my variables?

Variable: is a quantity, quality, or property that you can measure.

Value: is the state of a variable when you measure it. The value of a variable may change from measurement to measurement.

Observation: is a set of measurements made under similar conditions

Tabular data: is a set of values, each associated with a variable and an observation.

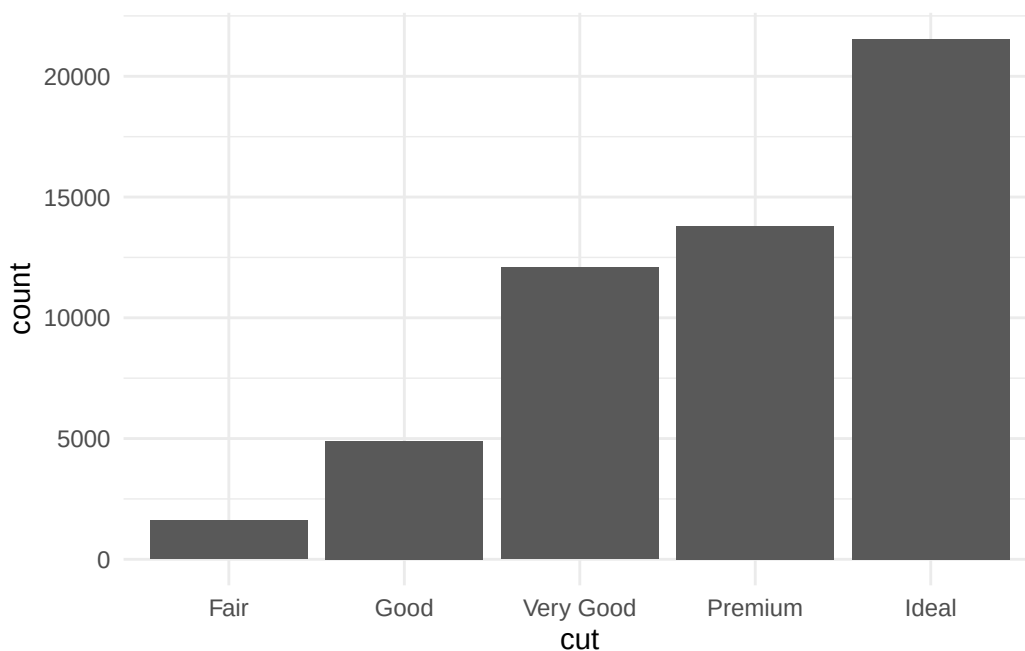
Variation

Variation is the tendency of the values of a variable to change from measurement to measurement.

Visualising Distributions

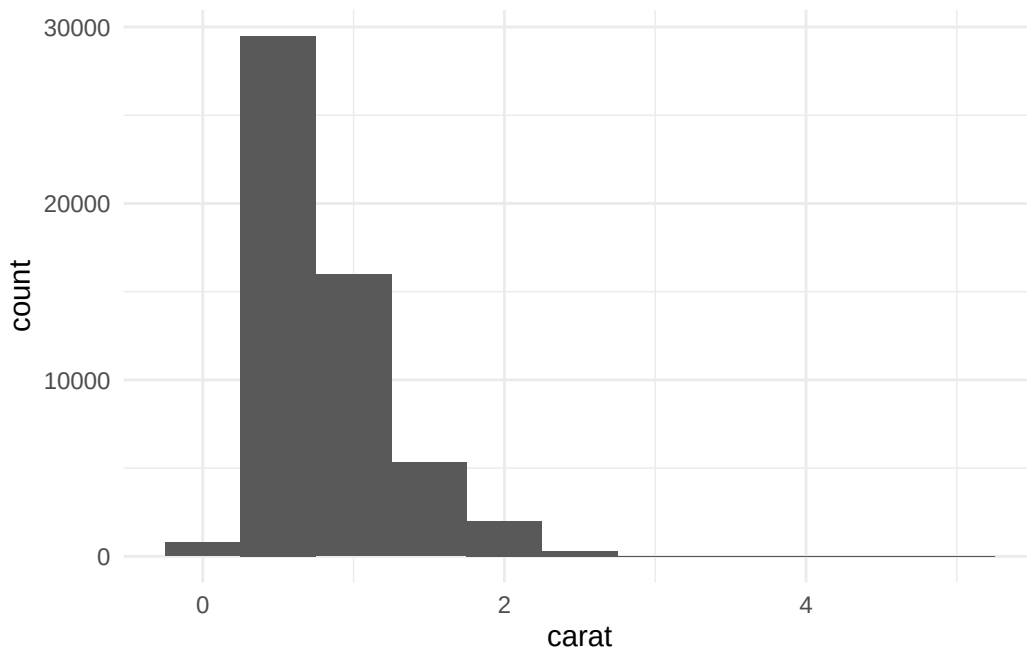
A variable is categorical if it can only take one of a small set of values. In R, categorical variables are usually saved as factors or character vectors. To examine the distribution of a categorical variable, use a bar chart:

```
ggplot(data = diamonds) +  
  geom_bar(mapping = aes(x = cut)) +  
  theme_minimal()
```



A variable is continuous if it can take any of an infinite set of ordered values. Numbers and date-times are two examples of continuous variables. To examine the distribution of a continuous variable, use a histogram:

```
ggplot(data = diamonds) +  
  geom_histogram(mapping = aes(x = carat), binwidth = 0.5) +  
  theme_minimal()
```



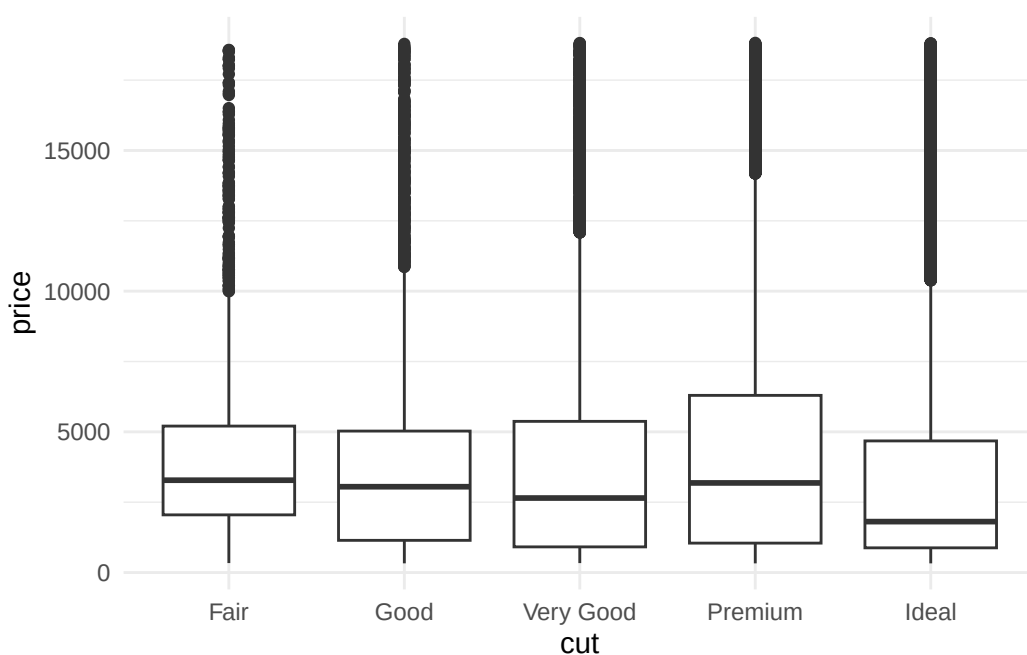
Covariation

Covariation is the tendency for the values of two or more variables to vary together in a related way. The best way to spot covariation is to visualise the relationship between two or more variables.

A categorical and continuous variable

One can use a box plot to explore this relationship.

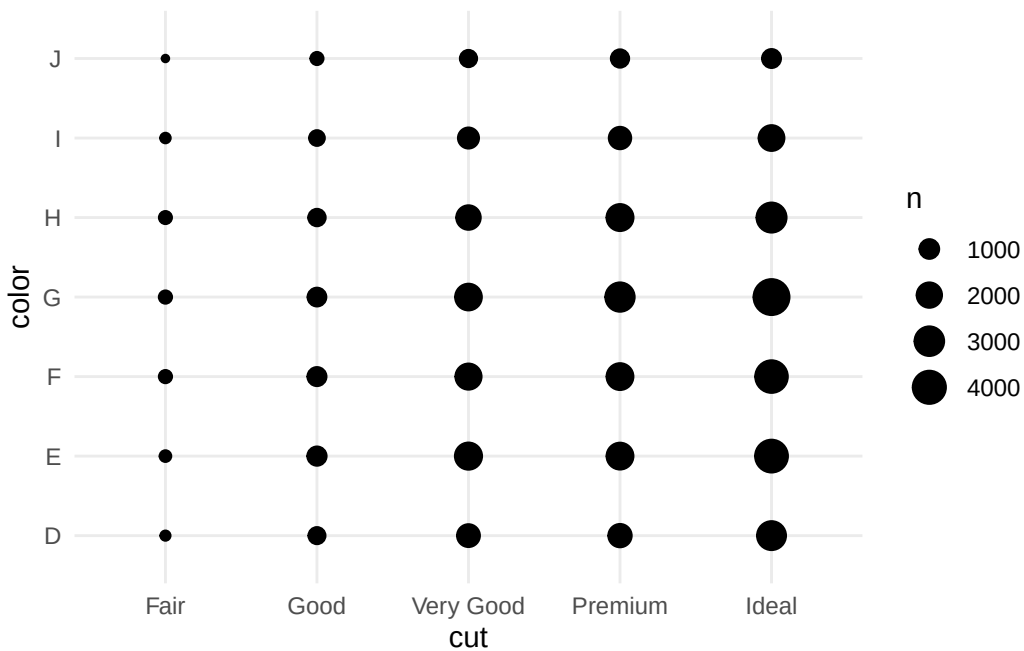
```
ggplot(data = diamonds, mapping = aes(x = cut, y = price)) +  
  geom_boxplot() + theme_minimal()
```



Two Categorical Variables

To visualise the covariation between categorical variables, you'll need to count the number of observations for each combination. One way to do that is to rely on the built-in `geom_count()`:

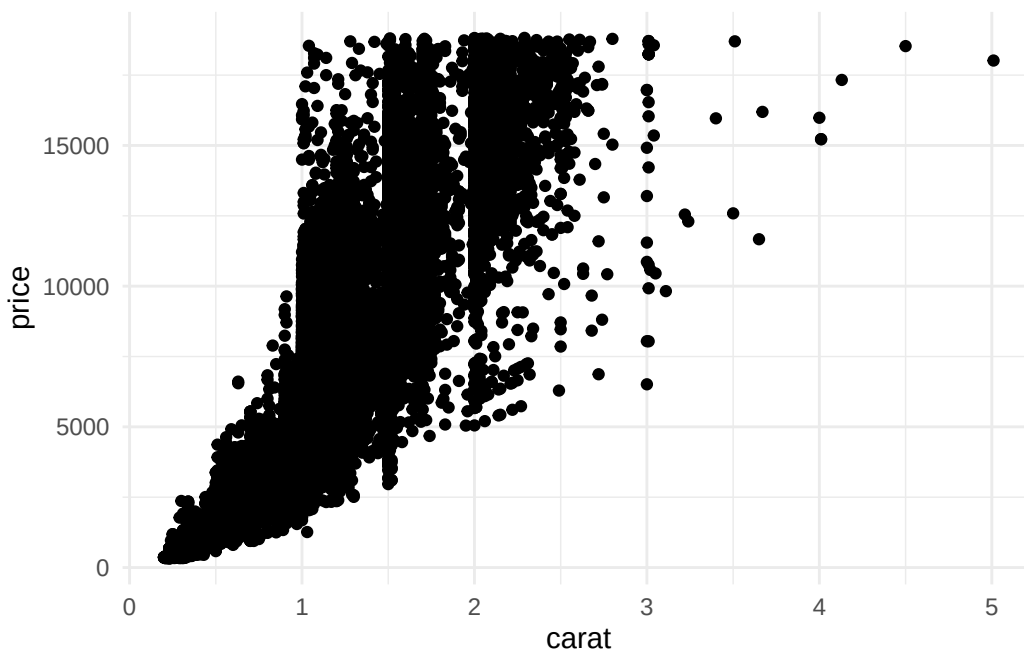
```
ggplot(data = diamonds) +  
  geom_count(mapping = aes(x = cut, y = color)) +  
  theme_minimal()
```



Two continuous variables

You've already seen one great way to visualise the covariation between two continuous variables through a scatter plot.

```
ggplot(data = diamonds) +  
  geom_point(mapping = aes(x = carat, y = price)) +  
  theme_minimal()
```



Importing Data Sets in R

Working with data provided by R packages is a great way to learn the tools of data science, but at some point you want to stop learning and start working with your own data.

We will explore how to import the following common types of files into R.

1. Excel Files
2. csv files
3. SPSS files
4. STATA files

Each of the above files has an equivalent package in R to import the respective file.

csv files

CSV files are the most commonly used data files. To import this we can use the two options namely:

- `read.csv()`
- `read_csv()` from the `readr` package

You have to specify the file path if you are not working inside a R Project.

Example is as shown:

```
mydata <- read.csv("/path to data file/data.csv")
```

The alternative form is given as: